

Bandwidth Throttling with NetEM



NetEM (Network Emulation) provides functionality for testing protocols, by emulating the network properties of wide-area networks. This article describes the use of NetEM in benchmarking the performance of a Web application, simulating Internet-like speeds on a LAN.

In a recent project, I was busy doing a Web application performance benchmark test. For most such benchmarks, it is important to simulate the typical end-user's average effective bandwidth. Since we used a fast network for our benchmarks, we had to find a way to somehow limit the bandwidth of our LAN so that it would resemble 'real' Internet throughputs. It took some research and reading to find the best way to do this. We found that there were commercial shaping products like Allot's Netenforcer (www.allot.com/html/products_netenforcer.shtm) and Packeteers Shaper (www.packeteer.com), but these were

costly. So instead of these devices/appliances, we decided to use the NetEM tool.

In our environment (see Figure 1), the NetEM box also plays the role of a Linux router. Recent Linux kernels have built-in network traffic shaping capabilities. Those capabilities, in combination with the command-line tool *tc* (a part of the *iproute2* package) can be used to set a bandwidth limit on one of your network interfaces, and even on incoming traffic on a specific port.

About traffic shaping

Traffic shaping is an attempt to control network traffic by prioritising network resources. It guarantees certain bandwidth,